

Introduction to Computer Organisation and Architecture UCCD1133



Chapter 3 Basic Concept of Logic

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Outline

- Boolean Algebra
- Boolean Analysis of logic circuit

Boolean Algebra

- Laws and theorems of Boolean Algebra.
- Apply these laws and theorems to:
 - Simplify expressions
 - Convert any Boolean expression into a sum-of-product (SOP) form
 - Convert non-canonical form to canonical form
- Boolean algebra is the mathematics of digital systems.
- Three terms that are used in Boolean algebra:
 - Variable: is a symbol (letter) used to represent logical quantity.
E.g. F, X, Y, Z and etc. Any variable can have a 1 or a 0 value.
 - Complement: is the inverse of variable.
E.g. complement of $A = \bar{A} = A'$

•

Fundamental of Boolean Algebra

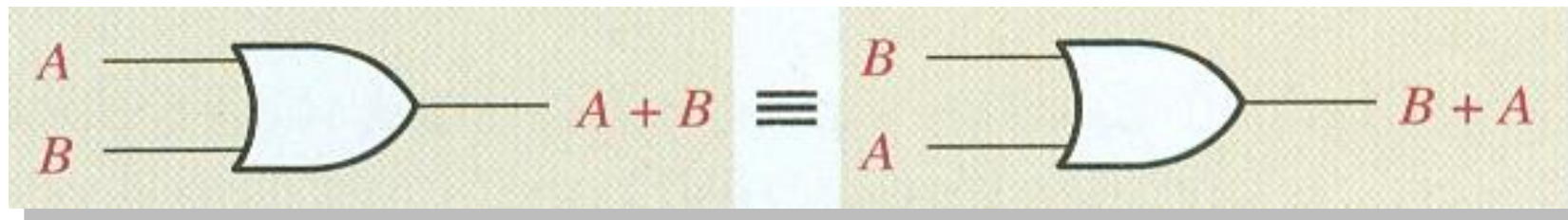
- Boolean algebra logical operators:
 - “ + ” => logical OR operator
 - “ . ” => logical AND operator
 - “ ’ ” or “ $\bar{}$ ” => logical NOT operator
 - E.g. on OR, AND and NOT computation

OR function with two variables = $x + y$	AND function with two variables = $x.y$	NOT function with a single variable = x'
$0 + 0 = 0$	$0.0 = 0$	$0' = 1$
$0 + 1 = 1$	$0.1 = 0$	$1' = 0$
$1 + 0 = 1$	$1.0 = 0$	
$1 + 1 = 1$	$1.1 = 1$	

Fundamental of Boolean Algebra

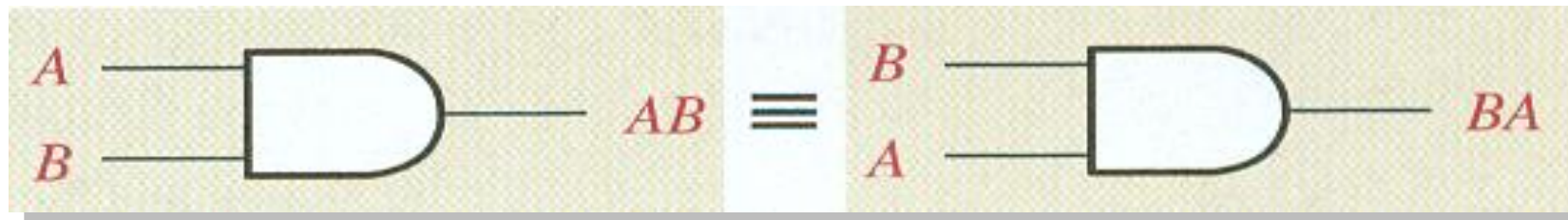
1. Commutative Laws (Order independent)

- **Commutative law of addition**



$$A + B = B + A$$

- **Commutative law of multiplication**

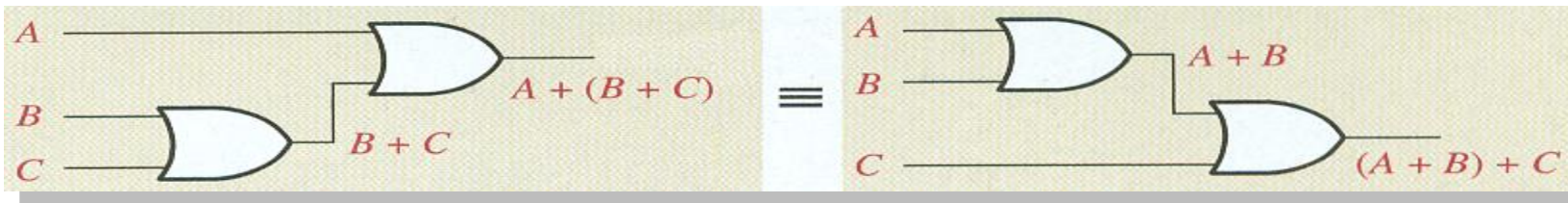


$$A * B = B * A$$

Fundamental of Boolean Algebra

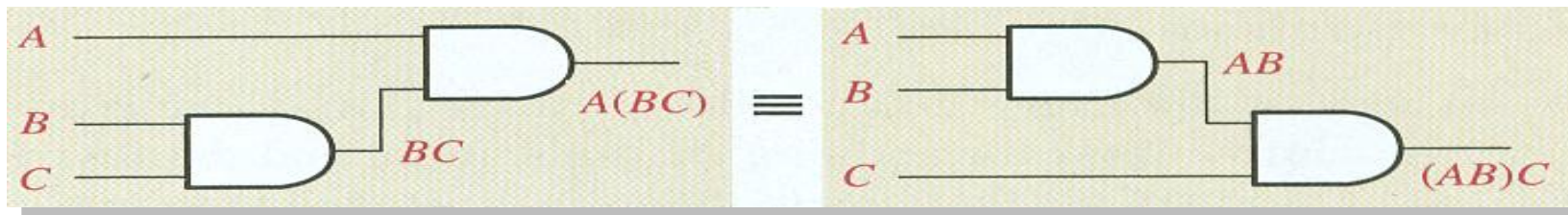
2. Associative Laws (grouping order-independent)

• Associative law of addition



$$A + (B + C) = (A + B) + C$$

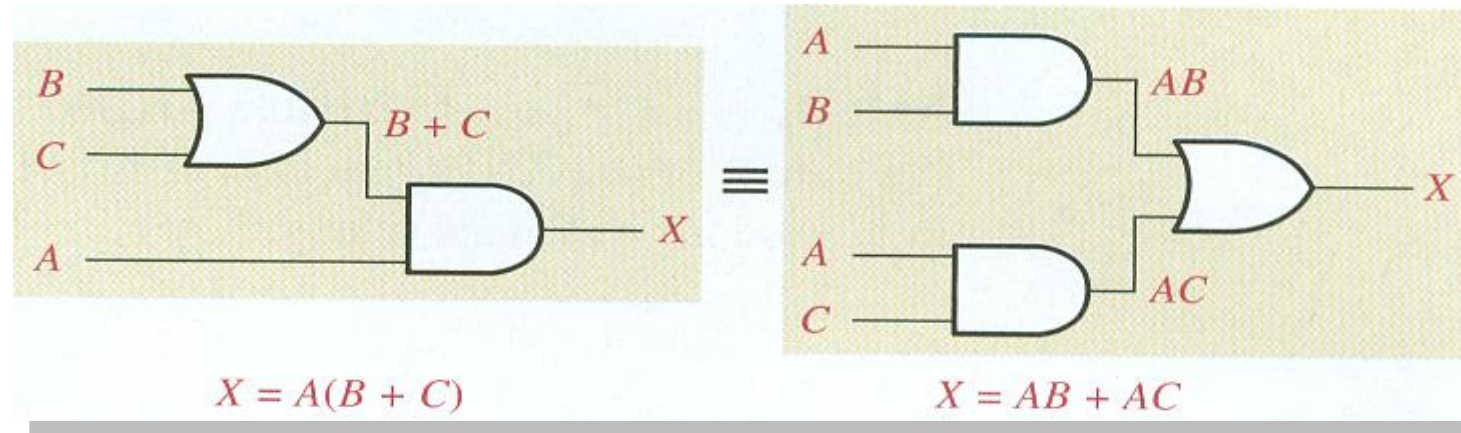
• Associative law of multiplication



$$A * (B * C) = (A * B) * C$$

Fundamental of Boolean Algebra

3. Distributive Law (Expanding an expression by multiplying term by term)



$$A(B+C) = AB + AC$$

Fundamental of Boolean Algebra

- Rule 1 through 9 can be viewed in terms of their application to logic gate.
- Rule 10 through 12 is derived in terms of simpler rules and laws previously discussed.

1. $A + 0 = A$

2. $A + 1 = 1$

3. $A \cdot 0 = 0$

4. $A \cdot 1 = A$

5. $A + A = A$

6. $A + \bar{A} = 1$

7. $A \cdot A = A$

8. $A \cdot \bar{A} = 0$

9. $\bar{\bar{A}} = A$

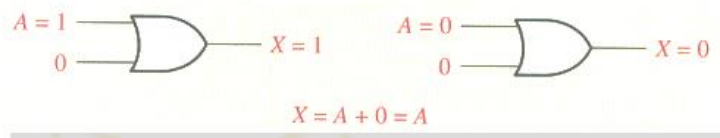
10. $A + AB = A$

11. $A + \bar{A}B = A + B$

12. $(A + B)(A + C) = A + BC$

Fundamental of Boolean Algebra

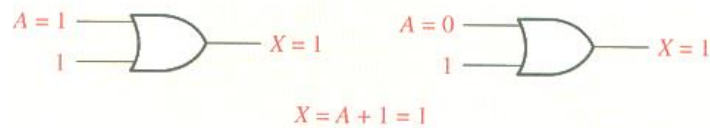
• Rule 1



A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

OR Truth Table

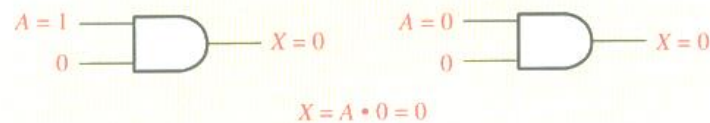
• Rule 2



A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

OR Truth Table

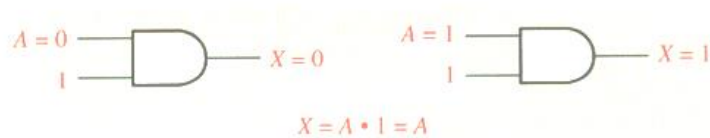
• Rule 3



A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

AND Truth Table

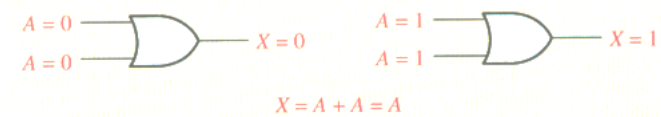
• Rule 4



A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

AND Truth Table

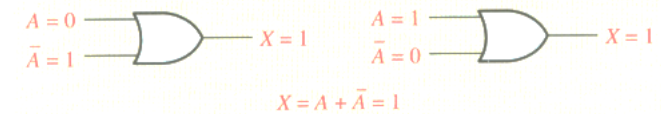
• Rule 5



A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

OR Truth Table

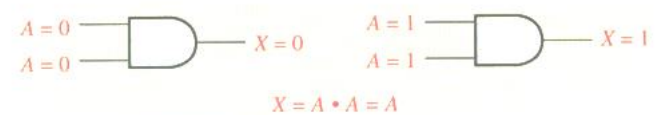
• Rule 6



A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

OR Truth Table

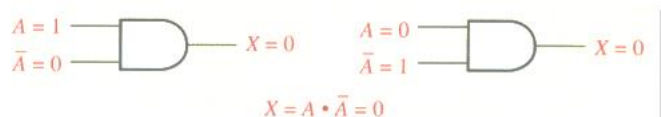
• Rule 7



A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

AND Truth Table

• Rule 8

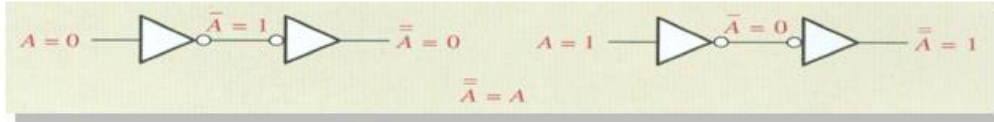


A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

AND Truth Table

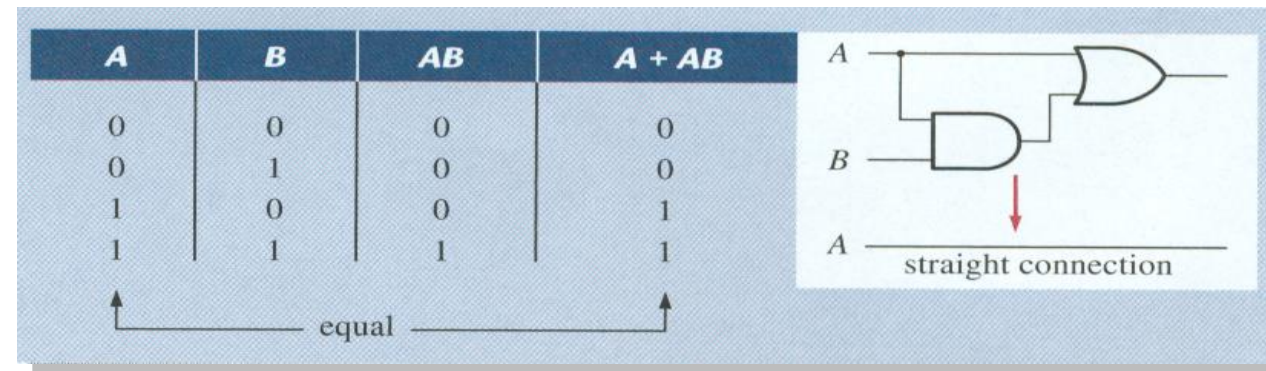
Fundamental of Boolean Algebra

- Rule 9



Rule 10: $A + AB$

$$\begin{aligned}
 A + AB &= A(1+B) && \text{Factoring (distributive law)} \\
 &= A(1) && \text{Rule 2: } 1 + B = 1 \\
 &= A && \text{Rule 4: } A \cdot 1 = A
 \end{aligned}$$



Fundamental of Boolean Algebra

- Rule 11: $A + \overline{A}B = A + B$

$$\begin{aligned}
 A + \overline{A}B &= A(1 + B) + \overline{A}B \\
 &= A + AB + \overline{A}B \\
 &= A + B(A + \overline{A}) \\
 &= A + B(1) \\
 &= A + B
 \end{aligned}$$

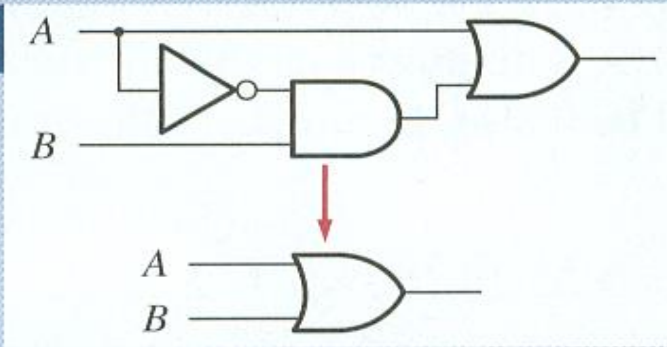
Rule 10: $A = A + AB$

Factoring (distributive law)

Rule 6: $A + \overline{A} = 1$

Rule 4: $B \cdot 1 = B$

A	B	$\overline{A}B$	$A + \overline{A}B$	$A + B$
0	0	0	0	0
0	1	1	1	1
1	0	0	1	1
1	1	0	1	1



↑ equal ↑

Fundamental of Boolean Algebra

- Rule 12: $(A+B)(A+C) = A+BC$

$$\begin{aligned}
 (A+B)(A+C) &= AA + AC + AB + BC \\
 &= A + AC + AB + BC \\
 &= A(1+C) + AB + BC \\
 &= A(1) + AB + BC \\
 &= A(1+B) + BC \\
 &= A + BC
 \end{aligned}$$

Factoring (distributive law)

Rule 7: $A \cdot A = A$

Factoring (distributive law)

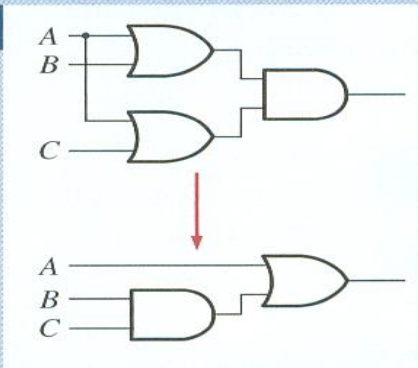
Rule 2: $C + 1 = 1$, Rule 4: $A \cdot 1 = A$

Factoring (distributive law)

Rule 2: $B + 1 = 1$, Rule 4: $A \cdot 1 = A$

A	B	C	A + B	A + C	(A + B)(A + C)	BC	A + BC
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	1	1	1	1
1	0	0	1	1	1	0	1
1	0	1	1	1	1	0	1
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1

↑ equal ↑



Fundamental of Boolean Algebra

DeMorgan's 1st theorem

- The complement of a product of variables is equal to the sum of the complemented variables.

$$\overline{AB} = \overline{A} + \overline{B}$$

- Applying DeMorgan's first theorem to gates:



NAND

Negative-OR

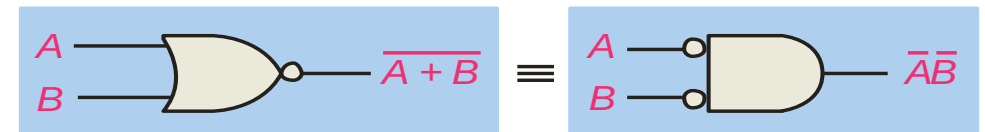
Inputs		Output	
A	B	$\overline{AB}$	$\overline{A} + \overline{B}$
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0

DeMorgan's 2nd theorem

- The complement of a sum of variables is equal to the product of the complemented variables.

$$\overline{A + B} = \overline{A} \overline{B}$$

- Applying DeMorgan's second theorem to gates:



NOR

Negative-AND

Inputs		Output	
A	B	$\overline{A + B}$	$\overline{A} \overline{B}$
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

Axioms of Boolean Algebra

- Boolean Algebra is based on a set of rules that are derived from a number of assumptions called axioms.
- Following are the summary of them with single variable

$$1a. \quad 0 \cdot 0 = 0$$

$$1b. \quad 1 + 1 = 1$$

$$2a. \quad 1 \cdot 1 = 1$$

$$2b. \quad 0 + 0 = 0$$

$$3a. \quad 0 \cdot 1 = 1 \cdot 0 = 0$$

$$3b. \quad 1 + 0 = 0 + 1 = 1$$

$$4a. \quad \text{If } x = 0, \text{ then } \bar{x} = 1$$

$$4b. \quad \text{If } x = 1, \text{ then } \bar{x} = 0$$

$$5a. \quad x \cdot 0 = 0$$

$$5b. \quad x + 1 = 1$$

$$6a. \quad x \cdot 1 = x$$

$$6b. \quad x + 0 = x$$

$$7a. \quad x \cdot x = x$$

$$7b. \quad x + x = x$$

$$8a. \quad x \cdot \bar{x} = 0$$

$$8b. \quad x + \bar{x} = 1$$

$$9a. \quad \bar{\bar{x}} = x$$

- Two & three variables theorem

$$10a. \quad x \cdot y = y \cdot x$$

Commutative

$$10b. \quad x + y = y + x$$

$$11a. \quad x \cdot (y \cdot z) = (x \cdot y) \cdot z$$

Associative

$$11b. \quad x + (y + z) = (x + y) + z$$

$$12a. \quad x \cdot (y + z) = x \cdot y + x \cdot z$$

Distributive

$$12b. \quad x + y \cdot z = (x + y) \cdot (x + z)$$

$$13a. \quad x + x \cdot y = x$$

Absorption

$$13b. \quad x \cdot (x + y) = x$$

$$14a. \quad x \cdot y + x \cdot \bar{y} = x$$

Combining

$$14b. \quad (x + y) \cdot (x + \bar{y}) = x$$

$$15a. \quad \overline{x \cdot y} = \bar{x} + \bar{y}$$

DeMorgan's theorem

$$15b. \quad \overline{x + y} = \bar{x} \cdot \bar{y}$$

$$16a. \quad x + \bar{x} \cdot y = x + y$$

$$16b. \quad x \cdot (\bar{x} + y) = x \cdot y$$

$$17a. \quad x \cdot y + y \cdot z + \bar{x} \cdot z = x \cdot y + \bar{x} \cdot z \quad \text{Consensus}$$

$$17b. \quad (x + y) \cdot (y + z) \cdot (\bar{x} + z) = (x + y) \cdot (\bar{x} + z)$$

Axioms of Boolean Algebra

Example 5: Find the simplest term

$$xy + x'z + yz$$

Solution:

$$\begin{aligned} xy + x'z + yz &= xy + x'z + 1yz \\ &= xy + x'z + (x + x')yz \\ &= xy + x'z + xyz + x'yz \\ &= (xy + xyz) + (x'z + x'yz) \\ &= xy + x'z \end{aligned}$$

Algebraic Representation of a Logic Function

$$\underbrace{f(A,B,C)}_{\text{Logic function of 3 variables}} = \underbrace{AB + A'C + AC'}_{\text{Logic expression}} \quad \left. \begin{array}{l} \text{Logic equation} \\ \text{variable} \\ \text{compliment} \end{array} \right\} \text{literal (e.g. 6 literals and 3 variables)}$$

- Algebra – used as a mathematical representation of a logic function
- Example: if $A = 1, B = 0, C = 0$, then

$$\begin{aligned} f(1,0,0) &= 1.0 + 1'.0 + 1.0' \\ &= 0 + 0 + 1 \\ &= 1 \end{aligned}$$

Boolean Algebra

Exercise

1. Apply DeMorgan's theorem to each of the following expressions

i. $\overline{ABC+DEF}$

ii. $\overline{A + \overline{BC} + CD + \overline{BC}}$

2. Simplify the following using Boolean algebra techniques and draw the simplified logics circuit:

i. $AB + AB'(C + BD)$

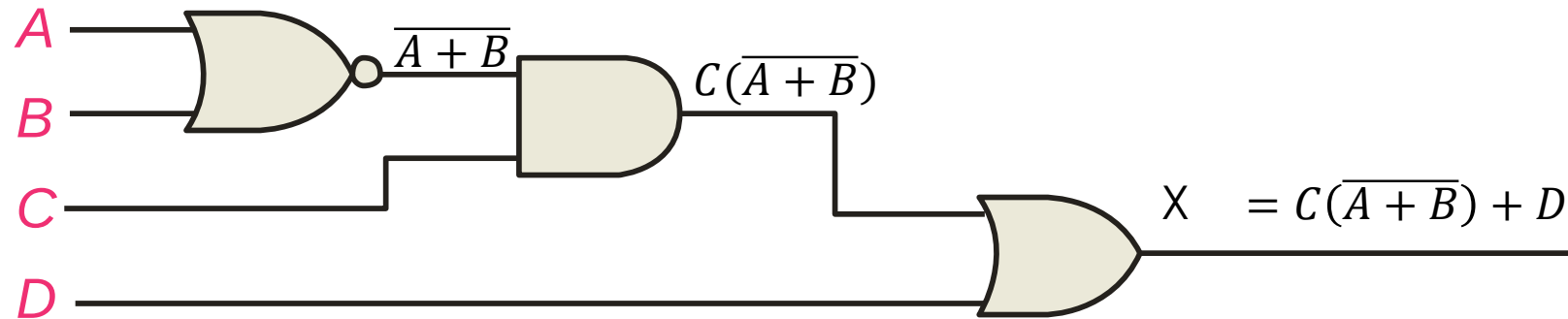
ii. $\overline{AB + AC} + A'B'C$

Boolean Analysis of a logic circuit

- A **combinational logic circuit** implements logical functions where its outputs depend only on its current combination of input values.
- Combinational logic circuits can be analyzed by writing the expression for each gate and combining the expressions according to the rules for Boolean algebra.
- Begin at the **left-most input** and work toward the **final output**

Boolean Analysis of a logic circuit

Example 10: Derive output expression, X .

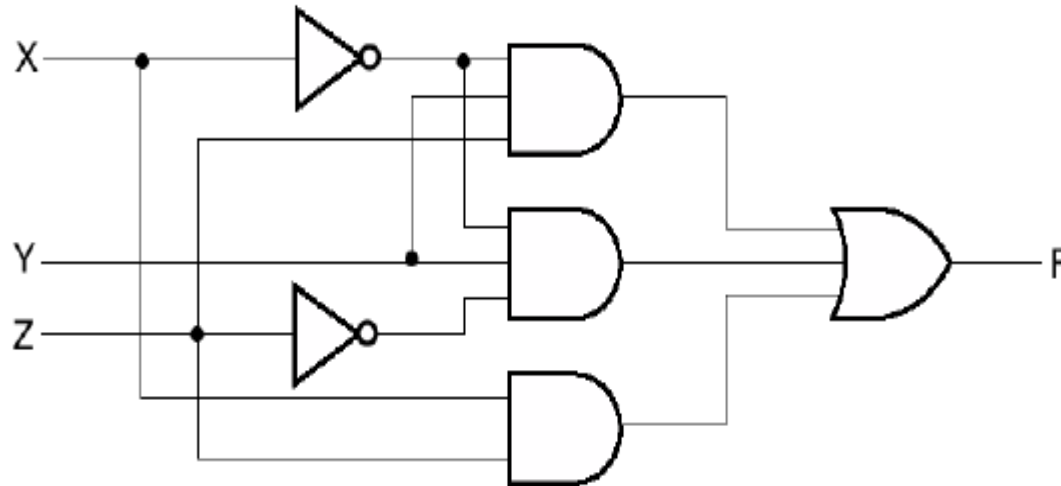


Boolean Simplification

- What is the primary motivation for using Boolean algebra to simplify logic expressions?
 - **It reduce a particular expression to its simplest form**
 - make it easier to understand the overall function of the circuit.
 - **Reduce number of inputs required**
 - efficient expression
 - **reduce the number of gates**
 - Compact design
 - Lower cost
 - Faster

Boolean Simplification

Example 11: Find the Boolean expression for given logics circuit. Then simplify the Boolean expression and illustrate the simplified circuit



Solution:

$$\mathbf{F = \bar{X}YZ + \bar{X}Y\bar{Z} + XZ}$$

Boolean Simplification

Do the simplification of Boolean expression as follows:

$$F = \bar{X}YZ + \bar{X}Y\bar{Z} + XZ$$

$$F = \bar{X}Y(Z + \bar{Z}) + XZ$$

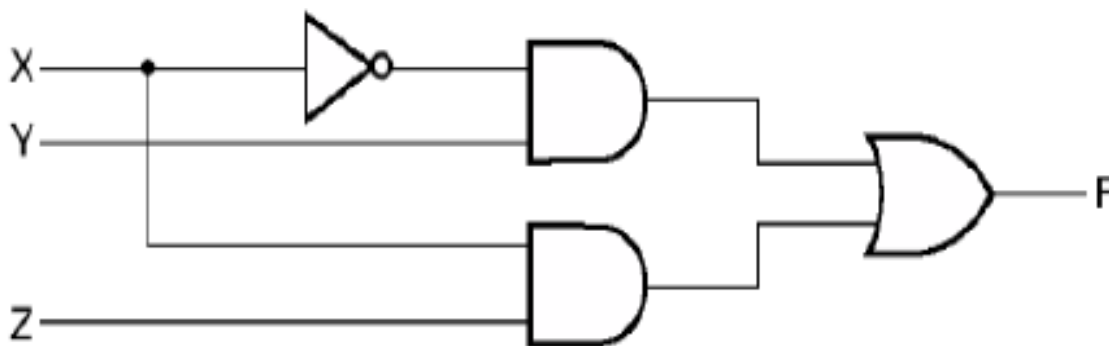
$$F = \bar{X}Y(1) + XZ$$

$$F = \bar{X}Y + XZ \quad \text{Simplified Boolean Expression}$$

Distributive law

Rule 6: $Z + Z' = 1$

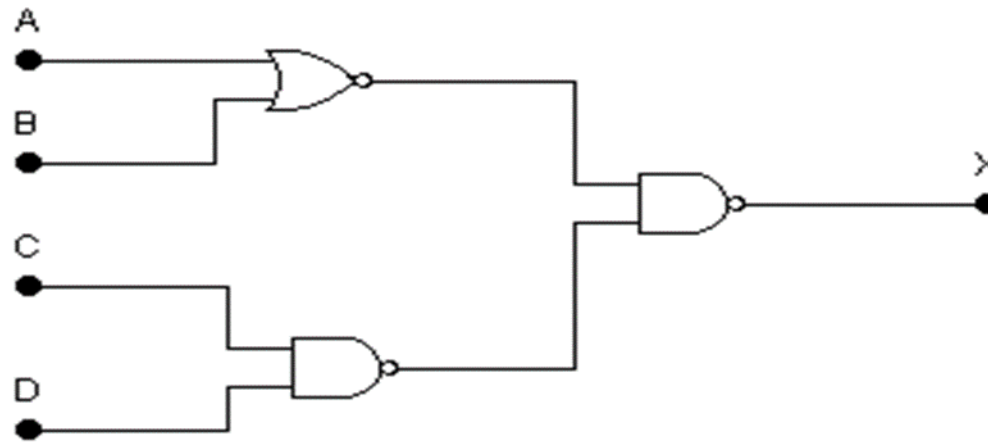
Rule 4: $X'Y \cdot 1 = X'Y$



Boolean Simplification

Exercise

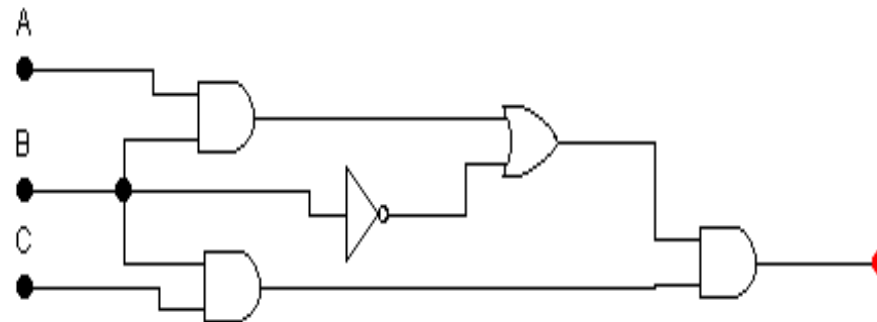
Write the Boolean expression of the following circuits and simply the circuits.



Constructing Truth Table basic on logic circuit – Method I

- Construct a truth table from a logic circuit
 1. List all the input variable combinations of 1 and 0 in binary sequentially
 2. Place the output logic for each combination of input
 3. Base on the result found write out the Boolean expression.

Example 12: Construct truth table for the specific logic gate

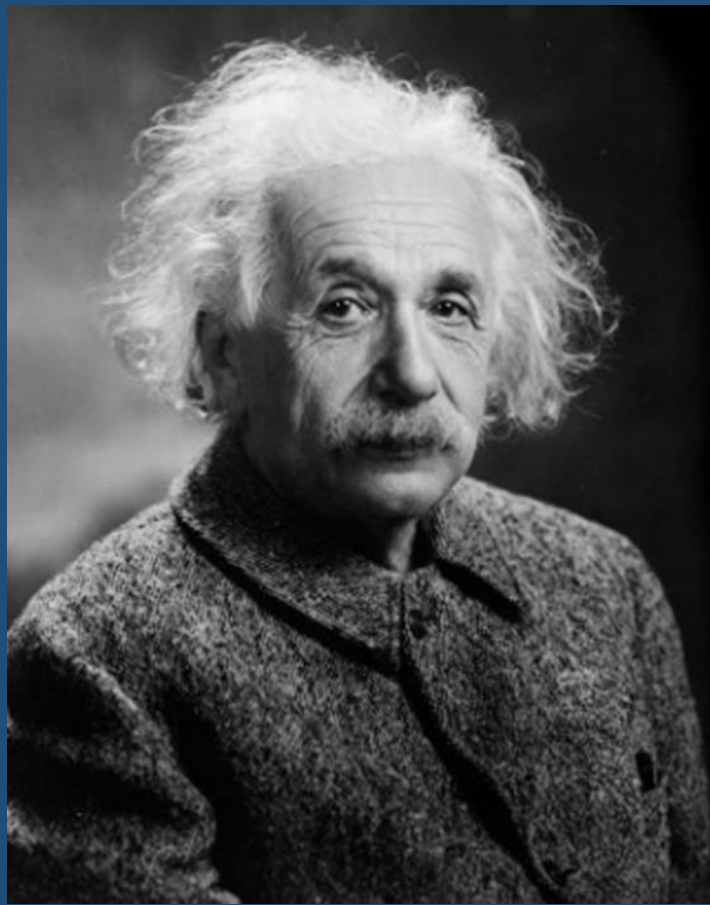


Constructing Truth Table basic on logic circuit – Method I

A	B	C	AB	$\overline{B}$	BC	$AB + \overline{B}$	$(AB + \overline{B}) BC$	Minterm
0	0	0	0	1	0	1	0	$\overline{A} \overline{B} \overline{C}$
0	0	1	0	1	0	1	0	$\overline{A} \overline{B} C$
0	1	0	0	0	0	0	0	$\overline{A} B \overline{C}$
0	1	1	0	0	1	0	0	$\overline{A} B C$
1	0	0	0	1	0	1	0	$A \overline{B} \overline{C}$
1	0	1	0	1	0	1	0	$A \overline{B} C$
1	1	0	1	0	0	1	0	$AB \overline{C}$
1	1	1	1	0	1	1	1	ABC

Limitations of Algebraic Method for Logic Optimisation

- Limitation of Boolean algebra method for logic optimization:
 1. Solution is not guaranteed to be minimum.
 - i. Non-systematic steps to reach a desired minimum solution.
 - ii. Approach is heuristic.
 - iii. Repeated search is based on intuition and experience.
 - iv. Time consuming and error-prone
 2. Impractical for large number of variables since relies heavily on the ability of the designer to use theorems and laws.
 - i. Slow and error-prone.
 - ii. Usage is limited to small number of variables.
 3. The expression is often made complex due to expansion before it can be simplified.
 4. Other better methods that can overcome the limitations of Boolean algebra method
 - i. Karnaugh Map method
 - ii. Quine-McCluskey method



“Creativity is
intelligence
having fun”
– Albert Einstein



Q & A
Thank you